

Maxime Meyer

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SUMMARY STATEMENT

PhD student in the Department of Mathematics at NUS focusing on approximation theory and the mathematical foundations of transformer models. My research develops a rigorous framework for understanding how architectural parameters shape expressiveness, providing theoretical principles for designing lighter, more efficient, and resource-aware AI systems.

EDUCATION

PhD in Mathematics

Feb. 2025 – Dec. 2028 (Expected)

National University of Singapore - GPA : 4.4/5.0

Singapore

- Supervised by Vincent Tan and Caroline Chaux
- Approximation Theory of Transformer Models

Master of Science by Research in Mathematics

Aug. 2023 – Jan. 2025

National University of Singapore - GPA : 4.3/5.0

Singapore

- Master Thesis with Patrick Rebentrost on Online Learning of Quantum States
- Pure Maths classes : Lie Theory, Representation Theory, Set Theory, Commutative and Homological Algebra

French Engineering School

Aug. 2021 – Jan. 2025

ENSTA Paris - GPA : 3.7/4.0

Saclay, France

- Equivalent of a Master of Science
- Specialized in Probabilities and Stochastic Calculus

Master Year 1 Applied Mathematics and Statistics

Aug. 2022 – Aug. 2023

Institut Polytechnique de Paris - GPA : 3.8/4.0

Saclay, France

Classes Préparatoires MPSI-MP*

Sep. 2019 – June 2021

Lycée Louis Le Grand - GPA : 4.0/4.0

Paris, France

- Equivalent of a Bachelor of Science in Maths and Physics

PUBLICATIONS

- [Maxime Meyer*](#), [Mario Michelessa*](#), [Caroline Chaux](#), and [Vincent Y. F. Tan](#) (2026). *How Many Different Outputs Can a Transformer Generate?* **ICML 2026 Oral**.
- [Dang Hoang Duy*](#), [Yannis Montreuil*](#), [Maxime Meyer*](#), [Axel Carlier](#), [Lai Xing Ng](#) and [Wei Tsang Ooi](#) (2026). *Online Learning-to-Defer with Varying Experts*. **AISTATS 2026**.
- [Maxime Meyer](#), [Mario Michelessa](#), [Caroline Chaux](#), and [Vincent Y. F. Tan](#) (2025). *Memory Limitations of Prompt Tuning in Transformers*. arXiv:2509.00421. Under Review
- [Maxime Meyer](#), [Soumik Adhikary](#), [Naixu Guo](#), and [Patrick Rebentrost](#) (2025). *Online Learning of Pure States is as Hard as Mixed States*. **NeurIPS'25**.

WORK EXPERIENCE

Research Assistant - Online Learning of Quantum States

Nov. 2024 – Jan. 2025

CQT, Centre for Quantum Technologies

Singapore

- Supervised by Patrick Rebentrost
- Studied the online learning setting for quantum tomography. Challenged the widely held assumption that pure states are easier to learn than mixed states and showed that both are equally complex in the online framework. Also introduced smoothed analysis in the online quantum state learning problem, which allows for an interpolation between the worst-case and the average-case scenarios.

Research Assistant - Computational Complexity

CQT, *Centre for Quantum Technologies*

Sep. 2024 – Oct. 2024

Singapore

- Supervised by Miklos Santha
- Analyzed the paper "The Complexity of Gradient Descent: $CLS = PPAD \cap PLS$ " and related problems

Research Intern - Monte Carlo Methods for Rare Probability Events

May 2023 – July 2023

Paris Saclay - SOLsTIS Research Team, INRAE

Saclay, France

- Supervised by Gabriel Lang and Pierre Gloaguen
- On the First Hitting Time Density of an Ornstein-Uhlenbeck Process
- Designed and implemented two algorithms estimating the hitting time of an Ornstein-Uhlenbeck Process (similar to a Brownian motion). Key challenges were a very long hitting time expectation (which makes standard Monte Carlo methods impractical), and explicit formulas that are particularly hard to estimate.

AWARDS AND SCHOLARSHIPS

AISG PhD Fellowship (SGD \$390,000)	2026
Department Graduate Teaching Award	2025
IPAL PhD Scholarship	2025

LANGUAGES

Coding

- Python
- C++
- R, Matlab, Prolog, C

Other

- French - native
- English - fluent
- German, Chinese - intermediate

SPORTIVE AND CULTURAL ACHIEVEMENTS

Head of ENSTA's boxing team

- Medals at the Regional University Championships of 2022 and 2023 and qualification for nationals.
- Organized a boxing event attended by a couple hundred spectators
- Coached in several associations

Vice Champion of Singapore in Sport Sambo

Participated in the 2024 Chessboxing World Championships in Armenia

Head of ENSTA's theatre company